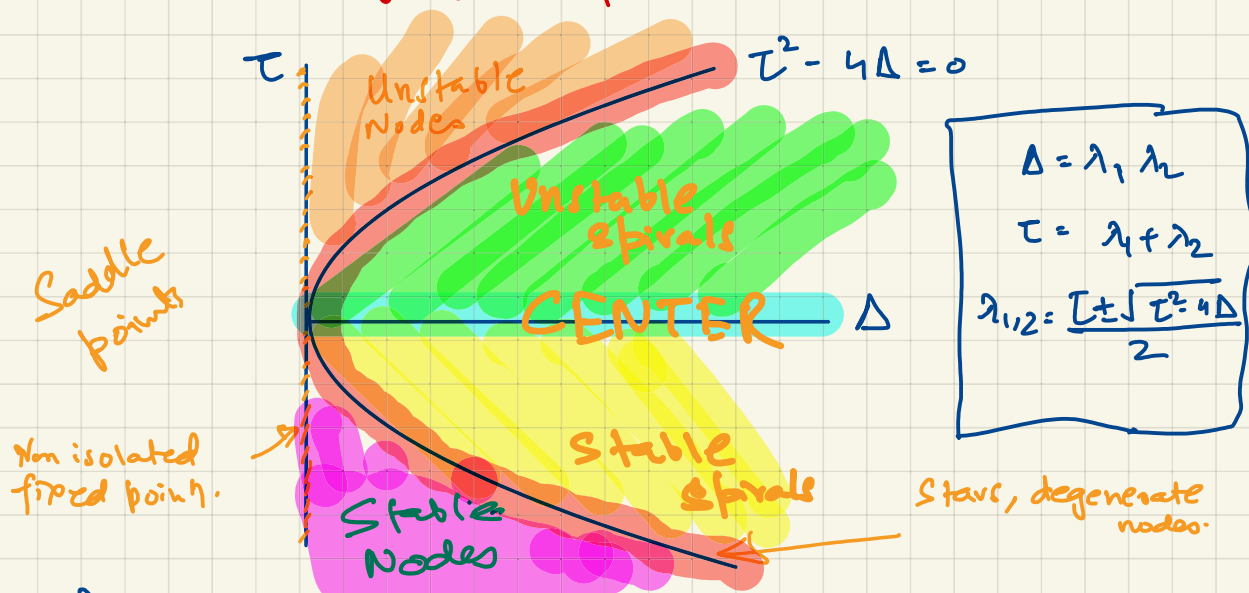


Classification of fixed points:



- 1) For $\Delta < 0$, eigenvalues are real and have opposite sign so they make a **saddle point**
 - 2) For $\Delta = 0, \tau = 0$, eigenvalues are always zero. This implies that the whole line of fixed points.
 - 3) For $\Delta = 0, \tau \neq 0$, at least one of the eigenvalues is zero and so origin is not an isolated fixed point and there is a whole line of fixed points.
 - 4) For $\Delta > 0, \tau^2 - 4\Delta \neq 0$, the eigenvalues are
 - a) Real with same sign creating **Nodes** if $\tau^2 - 4\Delta > 0$,
 - If $\tau < 0$ then the eigenvalues are negative and so a **stable node** is created.
 - If $\tau > 0$ then the eigenvalues are positive and so a **unstable node** is created.
 - b) Complex conjugate creating **spirals and centers** if $\tau^2 - 4\Delta < 0$
 - If $\tau < 0$ then $\text{Re}(\lambda_i) < 0$ and so it generates a **stable spiral** with a **stable fixed point**.
 - If $\tau > 0$ then $\text{Re}(\lambda_i) > 0$ and so it generates a **unstable spiral** with a **unstable fixed point**.
 - If $\tau = 0$ then $\text{Re}(\lambda_i) = 0$ and so it generates a **center** with a origin which is **Neutrally stable**.
 - 5) **Borderline cases:** $\tau^2 - 4\Delta = 0$, the eigenvalues are equal.
 - a) If there are two independent eigenvectors then it creates a **star node**.
 - b) If there is one eigenvector then it creates a **degenerate node**.
- Borderline cases:** Centers, degenerate nodes, stars, non isolated fixed points.